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☐ ٤أ

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6

* رقم الكمبيوتر



7

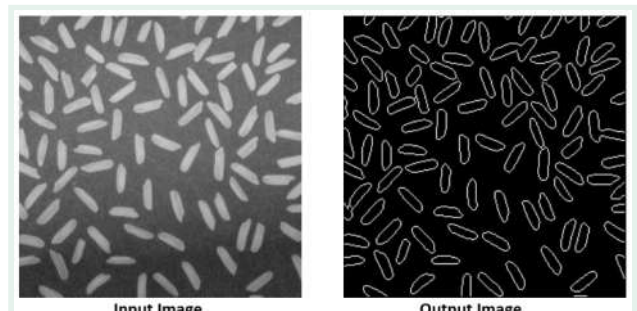
* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)



8

For the given following Figure:
Choose the suitable operation

(1 Point)



Binarization
Edge extraction

Object segmentation

Sharpening

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The general form of log transformations is _____ (1 Point)

$s = c + \log(1 + r)$

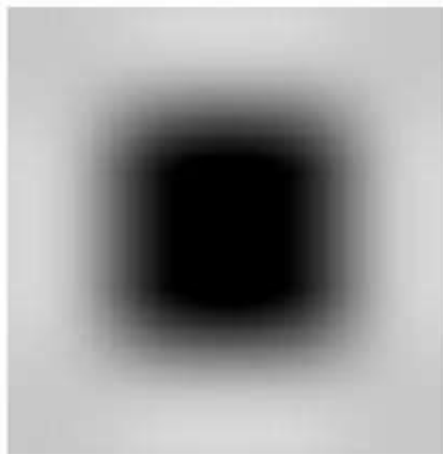
$s = c \cdot \log(1 - r)$

$s = c - \log(1 - r)$

$s = c \cdot \log(1 + r)$

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For the given following Figure: choose the suitable operation. (1 Point)



additive image offset

additive noise

blending

subtractive image offset

negative of an image

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Suppose that a 3-bit image has the intensity distribution, where the number of pixels that have intensity are given in the following Table:

The computed values of third row

are..... (2 Points)

r_k	$r_0 = 0$	$r_1 = 1$	$r_2 = 2$	$r_3 = 3$	$r_4 = 4$	$r_5 = 5$	$r_6 = 6$	
n_k	10	8	9	2	14	1	5	
$p_r(r_k)$								
s_k								

0.2 0.16 0.18 0.04 0.28 0.02 0.1 0.02

10 8 9 2 14 1 5 1

10 18 27 29 43 44 49 50

0.2 0.36 0.54 0.58 0.86 0.88 0.98 1

1.4 1.12 1.26 0.28 1.96 0.14 0.7 0.14

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Given the following Table:

The computed values of fourth row

are..... (2 Points)

r_k	$r_0 = 0$	$r_1 = 1$	$r_2 = 2$	$r_3 = 3$	$r_4 = 4$	$r_5 = 5$	$r_6 = 6$	
n_k	10	8	9	2	14	1	5	
$p_r(r_k)$								
s_k								

0.2 0.16 0.18 0.04 0.28 0.02 0.1 0.02

1.4 2.52 3.78 4.06 6.02 6.16 6.86 7

1 3 4 4 6 6 7 7

1 2 4 4 6 6 7 7

1 3 4 5 6 6 7 7

1 1 1 0 2 0 1 0

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The convolution mask for a 3×3 mean filter is given by (1 Point)

$$\begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$$

True

False

14

a 1024-level image is called aimage. (1 Point)

8-bit

9-bit

10-bit

11-bit

15

Sum of all components of a normalized histogram is equal to (1 Point)

0

1

2

None of the mentioned

16

Negative of the image having intensity values $[0, L-1]$ is expressed by (1 Point)

$s = L-1$

$s = L-1-r$

$s = L+1-r$

$s = L-r$

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A low contrast image will have what kind of histogram? (1 Point)

The histogram that are concentrated on the dark side of gray scale

The histogram whose component are biased toward high side of gray scale

The histogram that is narrow and centered toward the middle of gray scale

The histogram that covers wide range of gray scale and the distribution of pixel is approximately uniform

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Which is first fundamental step in image processing? (1 Point)

filtration

image acquisition

image enhancement

image restoration

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For the given following Figure: choose the suitable operation on the output image. (1 Point)

input



filter

0	0	0
0	2	0
0	0	0

$-\frac{1}{9}$

1	1	1
1	1	1
1	1	1

c



- ☒ unchanged
- ☐ shift to left
- ☐ deblurring
- ☐ blurring
- ☐ sharpening

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An effective image encryption algorithm should have the ability to reduce the correlations between adjacent pixels of plain image. (1 Point)

- ☒ True
- ☐ False

21

The cipher image converted into a non-readable form after encryption is called a plain image. (1 Point)

- ☒ True
- ☐ False

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The output of a smoothing, linear spatial filtering is a _____ of the pixels contained in the neighbourhood of the filter mask. (1 Point)

☒ Average

☐ Dot product

☐ Sum

☐ Product

☐ Median

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A digital image is a visual representation of an object, a person, or a scene produced by an optical device such as a mirror, a lens, or a camera. (1 Point)

☒ True

☐ False

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Which of the following method is/are used for padding the image? (1 Point)

☒ Adding rows and column of 0 or other constant gray level

☐ Simply replicating the rows or columns

☐ All of the above

☒ None of the above

25

Method in which images are input and attributes are output is called (1 Point)

- ☒ low level processes
- ☐ edge level processes
- ☐ high level processes
- ☐ mid level processes

26

Information entropy equal to 8, for a 512-gray level image. (1 Point)

- ☒ True
- ☐ False

27

The value of BER is between 0 and 1. The larger the value, the better the robustness of the algorithm. (1 Point)

- ☒ True
- ☐ False

28

A neighborhood averaging filter in which all coefficients are equal is also referred to as a box filter (1 Point)

☒ True

☐ False

29

Imperceptibility, security, capacity, and robustness are among the many aspects of image encryption design. (1 Point)

☒ True

☐ False

30

Histogram is the technique processed in..... (1 Point)

☒ intensity domain

☐ frequency domain

☐ undefined domain

☐ spatial domain

31

Smallest possible neighbourhood in an image must be of size (1 Point)

☒ 4x4

☐ 3x3

☐ 2x2

☒ 1x1

32

In image we notice that the components of histogram are concentrated on the low side on intensity scale. (1 Point)

☒ bright

☐ colourful

☐ dark

☐ all of the mentioned

33

..... is example of Low level of image processing operations (1 Point)

☒ edges extraction

☐ regions

contrast enhancement
extraction of attributes

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Which of the following involves reversing the intensity levels of an image?
(1 Point)

Piecewise Linear Transformations

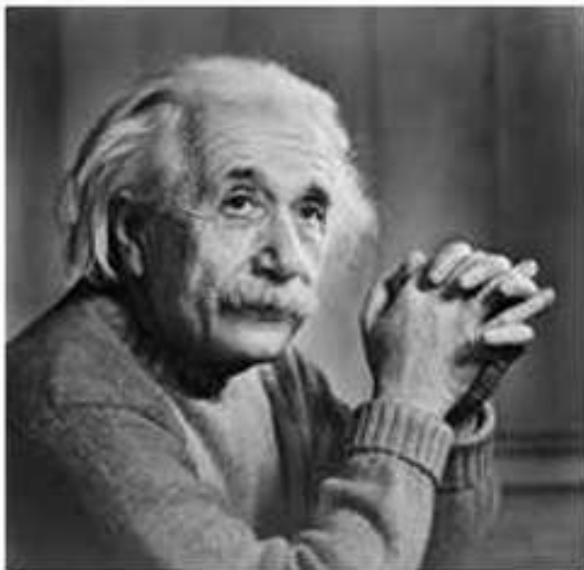
Log Transformations

Negative Transformations

Power Law Transformations

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Given the following Figure: Choose the suitable transformation. (1 Point)



Input Image



Output Image

Piecewise Linear Transformations



Log Transformations

Power Law Transformations

Negative Transformations

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